

Retained-Core with Conditional Context Hypothesis

Pressure-Tested Working Model for Transition Reliability

Working packet • pressure-tested hypothesis • bounded research note

Executive summary

Best working architecture: $\text{Lambda}_t = \text{sigma}(\text{U}_t + \text{C}_t)$

Universal retained core: $\text{U}_t = z(|d_t|) - 1.5 z(\text{rho}_t)$

Context is conditional: unnecessary in simpler regimes, required in metastable, folding-like, or excitable regimes.

Abstract

These experiments support a reliability-layer view of dynamics in which stable realization is not fully determined by visible state alone. The best-performing working model is a bounded reliability field:

$$\text{Lambda_t} = \text{sigma}(\text{U_t} + \text{C_t})$$

where U_t is a universal retained core and C_t is a conditional context field that becomes important in path-dependent regimes. A working empirical form for the universal core is:

$$\text{U_t} = \text{z}(|\text{d_t}|) - 1.5 \text{ z}(\text{rho_t})$$

Across the tested regimes, U_t was sufficient in simpler dynamics, while branch-stability or excitable-memory context was required in metastable, folding-like, or pulse-like regimes.

Core Claim

Transition reliability is governed by a universal retained core, with additional context fields required only in regimes where path dependence, metastability, branch instability, or excitable memory materially affect which futures are stably reachable.

Model

Proposed transition:

$$\text{G}\sim\text{_(t+1)} = \text{F}(\text{G_t}, \text{R_t})$$

Universal retained core:

$$\text{U_t} = \text{z}(|\text{d_t}|) - 1.5 \text{ z}(\text{rho_t})$$

Reliability field:

$$\text{Lambda_t} = \text{sigma}(\text{U_t} + \text{C_t})$$

Interpretation:

$$\text{P}(\text{stable realization of } \text{G}\sim\text{_(t+1)} \mid \text{G_t}, \text{R_t}, \text{C_t}) = \text{Lambda_t}$$

Interpretation of Terms

d_t behaves like retained directional structure strength. Its sign is not universal, but its magnitude repeatedly carried reliability signal.

rho_t behaves like retained destabilization burden. Stronger rho_t generally reduced reliability, and its weight repeatedly settled above 1, with 1.5 the best tested working value.

C_t is not arbitrary context. It is the extra information needed when the same apparent present state can still belong to different future-stability classes.

Regime Summary

Regime A-like: best described by $\Lambda_t \approx \sigma(U_t)$. Context was unnecessary and often harmful.

Regime B/C-like: best described by $\Lambda_t \approx \sigma(U_t + B_t - P_t)$, where B_t is branch stability and P_t is switching, oscillation, or over-commitment penalty.

Folding-path regime: after calibrating the success criterion, the richer branch context restored a strong monotone reliability ordering, while core-only was weak.

Out-of-family pulse regime: core-only already worked somewhat, but a light pulse/refractory context improved separation further.

What Was Falsified

- Direct retained-state forcing.
- Sign-aware forcing laws.
- Hard accept/reject transition laws.
- One universal scalar without context.
- Current attempts to derive a universal activation law from local features or nearest-neighbor divergence.

What Survived

- Visible-state-only descriptions are incomplete in the tested systems.
- Retained-state variables improve transition-reliability separation.
- Retained state belongs more naturally in a reliability/readiness layer than in a force law.
- The universal retained core transfers across multiple regimes.
- Context is conditional and becomes necessary in branch-structured or excitable regimes.

Best Current Formal Hypothesis

For a class of dynamical systems, transition reliability is better represented by a bounded retained reliability field $\Lambda_t = \sigma(U_t + C_t)$, where U_t is a universal retained core derived from retained directional structure and retained destabilization, and C_t is a context field required only in regimes with meaningful path dependence or instability memory.

Open Problem

We do not yet have a successful derived law for when C_t should activate strongly enough to replace hand-switched regime context.

Compact One-Line Conclusion

Transition reliability appears to be governed by a universal retained core, with additional context fields activated only when the regime has meaningful path dependence, branch instability, metastability, or excitable memory.